

Singular Value Decomposition (SVD)

Let $T \in L(V)$. The singular values of T are the non-negative sq. roots of the eigenvalues of T^*T , w/ each ^{distinct} eigenvalue λ repeated $\dim E(\lambda, T^*T)$ times. That is, if

$$T^*T \bar{v}_j = \lambda_j \bar{v}_j, \quad j \in \{1, \dots, n\} \text{ and } \dim V = n,$$

$\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n}$ are singular values of T .

Problem! Show that the singular values of T are the eigenvalues of $\sqrt{T^*T}$, with each eigenvalue λ repeated $\dim E(\lambda, \sqrt{T^*T})$ times.

Example: Define $T \in L(F^4)$ by

$$T(z_1, z_2, z_3, z_4) = (0, 3z_1, 2z_2, -3z_4)$$

$$\Rightarrow \langle \bar{z}, T^* \bar{w} \rangle = \langle T \bar{z}, \bar{w} \rangle$$

$$= 3z_1 \bar{w}_2 + 2z_2 \bar{w}_3 - 3z_4 \bar{w}_4$$

$$= \langle \bar{z}, (3\bar{w}_2, 2\bar{w}_3, 0, -3\bar{w}_4) \rangle$$

$$\Rightarrow T^* \bar{w} = (3w_2, 2w_3, 0, -3w_4)$$

$$T^*T(z) = T^*(0, 3z_1, 2z_2, -3z_4)$$

$$= (9z_1, 4z_2, 0, 9z_4)$$

$\{3, 3, 2, 0\}$ singular values

For std. basis $\mathcal{B}_V = \{\bar{e}_1, \bar{e}_2, \bar{e}_3, \bar{e}_4\}$

$$[T^*T]_{\mathcal{B}_V} = \begin{bmatrix} 9 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 9 \end{bmatrix}$$

$E(9, T^*T) = \text{span}(\bar{e}_1, \bar{e}_4)$
 $E(4, T^*T) = \text{span}(\bar{e}_2)$
 $E(0, T^*T) = \text{span}(\bar{e}_3)$

Thm. (Singular value decomposition)

Let $T \in L(V)$ has singular values $s_1, \dots, s_n$.
Then $\exists$ ~~an ONB~~ orthonormal bases

$\bar{e}_1, \dots, \bar{e}_n$ & $\bar{f}_1, \dots, \bar{f}_n$ of V s.t.

$$T\bar{e}_j = s_j \bar{f}_j, \quad j \in \{1, \dots, n\},$$

or, equivalently,

$$T\bar{v} = \sum_j s_j \langle \bar{v}, \bar{e}_j \rangle \bar{f}_j \quad \forall \bar{v} \in V.$$

Proof: $\dim V = n$. $T^*T \geq 0$, then from spectral

thm. $\exists$ an ONB $\bar{e}_1, \dots, \bar{e}_n$ of V w/

$$T^*T \bar{e}_k = \alpha_k \bar{e}_k, \quad k \in \{1, \dots, n\}, \quad s_k = \sqrt{\alpha_k}.$$

For each $k \in \{1, \dots, n\}$, let $\bar{f}_k = \frac{1}{s_k} T \bar{e}_k$.

If $j, k \in \{1, \dots, n\}$, then

$$\begin{aligned} \langle \bar{f}_j, \bar{f}_k \rangle &= \frac{1}{s_j s_k} \langle T \bar{e}_j, T \bar{e}_k \rangle = \frac{1}{s_j s_k} \langle T^* T \bar{e}_j, \bar{e}_k \rangle \\ &= \frac{1}{s_j s_k} \langle \alpha_j \bar{e}_j, \bar{e}_k \rangle = \frac{s_j}{s_k} \langle \bar{e}_j, \bar{e}_k \rangle = \delta_{jk} \\ &= \frac{s_j}{s_k} \delta_{jk} = \begin{cases} 0 & \text{if } j \neq k \\ 1 & \text{if } j = k \end{cases} \end{aligned}$$

$\therefore \bar{f}_1, \dots, \bar{f}_n$ is an ONB of V .

$$\text{for } \bar{v} \in V, \quad T\bar{v} = T \sum_j \langle \bar{v}, \bar{e}_j \rangle \bar{e}_j = \sum_j s_j \langle \bar{v}, \bar{e}_j \rangle \bar{f}_j.$$

Remark: $T\bar{e}_j = s_j \bar{f}_j$ for each $j \in \{1, \dots, n\}$,

$$[T]_{\mathcal{B}_V, \mathcal{B}_V'} = \begin{bmatrix} s_1 & & 0 \\ & \ddots & \\ 0 & & s_n \end{bmatrix}, \quad \mathcal{B}_V = \{\bar{e}_1, \dots, \bar{e}_n\}, \quad \mathcal{B}_V' = \{\bar{f}_1, \dots, \bar{f}_n\}.$$

Every operator on V has a diagonal matrix rep. w/ some orthonormal bases $\mathcal{B}_V, \mathcal{B}_V'$.
SVD holds also for $T \in L(V, W)$.

Thm. SVD for matrices

$A_{m \times n}$ is a matrix w/ singular values

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0 \text{ \& } \sigma_{r+1} = \sigma_{r+2} = \dots = \sigma_n = 0.$$

Then $\exists$ two lists of orthonormal vectors
 $\bar{u}_1, \dots, \bar{u}_m \in F^m$ \& $\bar{v}_1, \dots, \bar{v}_n \in F^n$ s.t.

$$A\bar{v}_i = \begin{cases} \sigma_i \bar{u}_i, & \text{for } i \in \{1, 2, \dots, r\} \\ 0 & \text{if } i \in \{r+1, \dots, n\} \end{cases}$$

or, equivalently,

$$A = U \Sigma V^* = \sum_{i=1}^r \sigma_i \bar{u}_i \bar{v}_i^*,$$

$$U = \begin{bmatrix} \bar{u}_1 & \dots & \bar{u}_m \\ | & & | \\ 1 & & 1 \end{bmatrix}_{m \times m}, \quad V = \begin{bmatrix} \bar{v}_1 & \dots & \bar{v}_n \\ | & & | \\ 1 & & 1 \end{bmatrix}_{n \times n},$$

$$\Sigma = \begin{bmatrix} \sigma_1 & & 0 & & 0 \\ & \ddots & & & \\ 0 & & \sigma_r & & 0 \\ \hline & & & 0 & \\ & & & & 0 \end{bmatrix}_{m \times n}$$

The singular values of A are sq. roots of eigenvalues of A^*A .

Example: Consider $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. $A^* A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Eigenvalues of $A^* A$ are $(\lambda_1, \lambda_2, \lambda_3) = (2, 1, 0)$

corresponding eigenvectors $\left(\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right)$
 $(2), (1), (0)$

After normalizing, $\bar{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, $\bar{v}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, $\bar{v}_3 = \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$.

$\{\bar{v}_1, \bar{v}_2, \bar{v}_3\} \rightarrow \text{ONB of } \mathbb{C}^3$.
 Singular values of A are $(\sigma_1, \sigma_2, \sigma_3) = (\sqrt{2}, 1, 0)$

$$V = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & -1/\sqrt{2} \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & 1 & 0 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} \sqrt{2} & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

To find $\bar{u}_1, \bar{u}_2$, we calculate

$$\bar{u}_1 = \frac{1}{\sigma_1} A \bar{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\bar{u}_2 = \frac{1}{\sigma_2} A \bar{v}_2 = \frac{1}{1} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$\{\bar{u}_1, \bar{u}_2\} \rightarrow \text{ONB of } \mathbb{C}^2$. $U = \begin{bmatrix} \bar{u}_1 & \bar{u}_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

SVD of A : $A = U \Sigma V^*$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{2} & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \\ -1/\sqrt{2} & 1/\sqrt{2} & 0 \end{bmatrix}$$